

MMI 803 · MATHEMATICAL PROBLEM SOLVING

MSc in Mathematical Innovation · Open University of Kenya

Module 5: Attacking Problems

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LEARNING OUTCOMES

1. Deploy specific “attack strategies” for unseen and unfamiliar problems.
2. Evaluate which strategy is most efficient and appropriate for a given problem type.
3. Apply working backwards, finding related problems, and problem simplification effectively.
4. Develop meta-cognitive awareness of *when to switch strategies*.

1 Module Overview

Having understood a problem (Module 4), the solver must *devise a plan* — Polya’s second phase. This module catalogues the primary attack strategies that experienced problem solvers deploy, with detailed worked examples for each. The key skill developed here is not merely knowing the strategies, but developing the *judgment* to select the right one and the *flexibility* to switch when the chosen strategy stalls.

THE EXPERT PROBLEM SOLVER

Research on expert mathematicians (Schoenfeld, 1985) shows that the primary difference between experts and novices is not that experts know more facts — it is that experts have a richer repertoire of *metacognitive strategies* and actively monitor their own progress, switching approaches when necessary.

2 Strategy 1: Working Backwards

When to use: When the desired *conclusion* or *end state* is clearer than the starting point.

WORKING BACKWARDS — Method

Start from the goal G . Ask: “What would need to be true one step before G ?” Repeat until you reach the given data D . Then reverse the chain to produce the forward proof.

Warning: Working backwards produces a *discovery path*, not necessarily a valid proof. Each backward step must be *reversible* (i.e., a logical equivalence, not merely an implication) for the forward argument to be valid.

EXAMPLE 1: Classic Working Backwards Proof

Problem: Prove that $\sqrt{2} + \sqrt{3} > \sqrt{10}$.

Working backwards:

$$\begin{aligned}
 &\sqrt{2} + \sqrt{3} > \sqrt{10} \\
 \iff &(\sqrt{2} + \sqrt{3})^2 > 10 \quad (\text{both sides positive, squaring is reversible}) \\
 \iff &2 + 2\sqrt{6} + 3 > 10 \\
 \iff &2\sqrt{6} > 5 \\
 \iff &4 \cdot 6 > 25 \quad (\text{squaring again, both sides positive}) \\
 \iff &24 > 25 \quad ???
 \end{aligned}$$

This is *false*. Conclusion: $\sqrt{2} + \sqrt{3} < \sqrt{10}$. The backwards process revealed the answer was opposite to our initial guess.

EXAMPLE 2: Working Backwards in a Planning Problem

Problem: A jug holds 8L. You have a 5L and a 3L container (no markings). Measure exactly 4L.

Goal: 4L in some container. Work backwards from this state.

Solution trace:

1. Fill 5L jug. Pour into 3L (fills it). 2L left in 5L jug.
2. Empty 3L jug. Pour 2L from 5L into 3L.
3. Fill 5L jug again. Pour from 5L into 3L until full (1L transferred). 4L remains in 5L jug. ✓

3 Strategy 2: Finding a Related (Simpler) Problem

When to use: When the problem is complex and a simpler version might yield insight into the solution structure.

RELATED PROBLEM — Method

Identify a simpler or more restricted version of the problem. Solve this version. Extract the *technique* or *pattern* from the simpler solution. Generalise to the original problem.

EXAMPLE 3: Handshake Problem via Related Problem

Original Problem: n people are at a party. Every person shakes hands with every other exactly once. How many handshakes total?

Simpler versions:

$n = 2$: 1 handshake

$n = 3$: 3 handshakes

$n = 4$: 6 handshakes

$n = 5$: 10 handshakes

Pattern: These are triangular numbers: $1, 3, 6, 10, \dots = \binom{n}{2}$.

General formula: $\binom{n}{2} = \frac{n(n-1)}{2}$ handshakes.

Proof: Each of the n people shakes hands with $n-1$ others, giving $n(n-1)$ *directed* handshakes. Since each handshake involves two people, divide by 2.

EXAMPLE 4: Analogy — Moving from 2D to 1D

Problem: Find the number of lattice paths from $(0,0)$ to (m,n) using unit steps right (R) or up (U).

Simpler 1D analogue: A path from 0 to m using only right steps — trivially 1 path. Not immediately helpful.

Reformulation: Every path consists of exactly m right steps and n up steps, in some order. Total steps: $m+n$. Choose which m of them are rightward:

$$\text{Number of paths} = \binom{m+n}{m} = \frac{(m+n)!}{m!n!}$$

4 Strategy 3: Problem Simplification

When to use: When the full problem is intractable but a simpler version is accessible.

Methods of simplification:

1. **Reduce variables:** Fix some variables to their simplest values.
2. **Reduce dimensions:** Solve the 2D version of a 3D problem.
3. **Reduce size:** Solve for $n = 3$ or $n = 4$ before n general.
4. **Remove constraints:** What if the problem had fewer conditions?

EXAMPLE 5: Simplification — Optimisation

Problem: Minimise $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 1$.

Simpler version: Minimise $f(x, y) = x^2 + y^2$ subject to $x + y = 1$.

By AM-QM inequality (or calculus): minimum is at $x = y = \frac{1}{2}$, giving $f = \frac{1}{2}$.

Generalise: For n variables summing to 1, minimum of $\sum x_i^2$ is $\frac{1}{n}$, achieved when all $x_i = \frac{1}{n}$ (by Cauchy-Schwarz or Lagrange multipliers).

5 Strategy 4: Introduce Auxiliary Elements

Introduce an additional object — a line, a point, a variable, a set — that does not appear in the original problem but whose introduction reveals structure.

EXAMPLE 6: Auxiliary Circle in Geometry

Problem: Prove that an angle inscribed in a semicircle is a right angle (Thales' Theorem).

Auxiliary element: Draw radius OB where O is the centre. Now triangles OAB and OBC are isosceles (since $OA = OB = OC = r$).

Let $\angle OAB = \alpha$ and $\angle OCB = \beta$. Then $\angle ABC = \alpha + \beta$. The angles of triangle ABC sum to 180: $\alpha + (\alpha + \beta) + \beta = 180 \Rightarrow 2(\alpha + \beta) = 180 \Rightarrow \angle ABC = 90$. $\square$

6 Strategy 5: Extreme Principle

When to use: When dealing with finite collections — consider the object with the *maximum* or *minimum* property. Its extremality often imposes useful constraints.

EXAMPLE 7: Extreme Principle

Problem: Prove that in any convex polygon, there exists a diagonal that lies entirely inside the polygon.

Extreme principle approach: Consider the *shortest diagonal*. Suppose it exits the polygon — then a shorter diagonal could be drawn inside the region it crossed, contradicting minimality. Hence the shortest diagonal lies inside.

7 Strategy 6: WLOG (Without Loss of Generality)

WLOG is used when a problem has a symmetry that allows us to assume a particular configuration, with the argument for all other configurations being identical by symmetry.

EXAMPLE 8: WLOG in Algebra

Problem: For real numbers $a \leq b$, prove $(a - b)^2 \geq 0$.

WLOG $a \leq b$ (the case $a > b$ is symmetric — just swap a and b ; $(b - a)^2 = (a - b)^2$).

Since $a \leq b$, we have $a - b \leq 0$. Squaring any real number gives a non-negative result.

□

8 Meta-Strategy: Monitoring and Switching

META-COGNITIVE PROTOCOL

As you work on a problem, ask at regular intervals (e.g., every 15 minutes):

1. *Am I making progress?* If not for 15 minutes, switch strategy.
2. *Is my current approach consistent with understanding the problem?*
3. *What have I learned from this path, even if it has not succeeded?*
4. *Have I considered all the strategies available?*

This monitoring prevents the most common failure mode: persistent application of a wrong strategy.

9 Strategy Selection Guide

Goal is clearer than start	Working backwards
Problem involves counting	Find a simpler related counting problem
Problem has many variables	Simplify (fix variables; reduce dimension)
Geometric problem	Introduce auxiliary point, line, or circle
Involves a finite set	Extreme principle (max/min element)
Problem is symmetric	Use WLOG
Pattern visible in small cases	Induction (Module 7)
Seems impossible	Seek invariant or Pigeonhole (Module 6)

10 Revision Questions

1. **[Working backwards]** Prove that $\sqrt{6} + \sqrt{7} > \sqrt{25 - \epsilon}$ for some specific $\epsilon > 0$ by working backwards and then writing the forward proof.
2. **[Related problem]** The problem of counting triangles in a triangulated polygon can be related to a simpler problem. Set up the recurrence and solve.
3. **[Simplification]** Solve the coin weighing puzzle for $n = 3$ coins (one fake, balance scale, 1 weighing). What insight does this give for the $n = 9$ case with 2 weighings?
4. **[Auxiliary elements]** In triangle ABC with $AB = AC$, D is the midpoint of BC . Prove $AD \perp BC$ using an auxiliary construction.
5. **[Extreme principle]** n points are placed in a unit square. Prove that at least two points are at distance $\leq \sqrt{2}/\lfloor \sqrt{n} \rfloor$ apart. (Hint: use a grid division.)
6. **[Meta-strategy]** Describe a problem-solving session where you got stuck. Identify which strategy you were using, why it stalled, and which strategy would have been more effective.
7. **[Strategy selection]** For each: identify the most appropriate strategy and begin the solution: (a) Find all $n \in \mathbb{N}$ such that $2^n + 1$ is prime; (b) Prove that any n points in the plane, not all collinear, determine at least n distinct lines.

MODULE 5 SUMMARY

- Attack strategies — working backwards, related problems, simplification, auxiliary elements, extreme principle, WLOG — are the core toolkit of the problem solver.
- Strategy selection is itself a skill: different problem types call for different approaches.
- Meta-cognitive monitoring — actively checking progress and switching strategies when stuck — distinguishes expert from novice problem solvers.
- Each failed strategy attempt is informative: it constrains the solution space.
- *There is no single path through every problem — the art is in choosing and navigating wisely.*

11 Further Reading

- **Polya, G.** (1945). *How to Solve It*. Part II: “Devising a Plan” and the heuristics dictionary.
- **Zeitz, P.** (2010). Chapter 3: “Strategies for Investigating Problems”.
- **Schoenfeld, A.H.** (1985). *Mathematical Problem Solving*. Academic Press.